TECHNICAL MEMO

TECHNICAL DOCUMENTATION
INFORMATION TO BE PROVIDED FOR ASSESSMENT

TYPE OF EVALUATION: INITIAL | MODIFICATION | EXTENSION | RENEWAL

REGULATION (EU) 2017/745

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→ RECOMMENDATION ON THE FORM OF THE TECHNICAL DOCUMENTATION SUBMITTED FOR ASSESSMENT

1 Approach

The Technical Documentation assessment requires a precise and comprehensive analysis of documents that may contain a significant number of pages in total. The quality of these files form (bookmarks, pagination, non-searchable PDFs, etc.) varies and impacts the internal organization.
The first part of this recommendation is a guide on the form of the Technical Documentation to be assessed and submitted to our departments. It therefore gives instructions for preparing the Technical Documentation solely in terms of their form.

This approach complies with the direction set out in the Regulation (EU) 2017/745 which specifies in the part of Annex II concerning the Technical Documentation: *"The technical documentation and, if applicable, the summary thereof to be drawn up by the manufacturer shall be presented in a clear, organized, readily searchable and unambiguous manner,* [...]."

Important note : As part of a conformity assessment on the basis of the EU type-examination according to Annex X of Regulation (EU) 2017/745, the Technical Documentation must be accompanied by a representative sample of the devices produced envisaged (referred to as "Type") as well as evidence demonstrating that the type was manufactured in accordance with the Technical Documentation concerned.

2 Technical Documentation structure

a → Composition

A Technical Documentation includes the following as a minimum a copy in electronic format (**electronic file**).

This file is essential for conducting the device conformity assessment. This file must be submitted in full to GMED at once. The submission of additional documents during the assessment (annexes, amendments, corrections, or otherwise) is not accepted and will not be taken into consideration.
The file must be written in either French or in English (English only for customers managed by GMED North America). GMED assessment report will be written in the same language as the Technical Documentation submitted.
In the event that the assessment process requires third party consultation (such as: Competent Authorities, EMEA, expert panel, etc.), the Technical Documentation relevant sections must be provided in the language required by the consulted parties.

Meanwhile, submission to GMED of the following additional information is highly recommended:

- Relevant pictures of the device and its commercial packaging, together with identification of labels positions and accompanying documents;
- Wherever possible, a sample in its commercial packaging (as it is intended to be put on the market), and clearly identified as such.

b → Consistency of the terminology

In general terms, the terminology used must be strictly consistent throughout the file.
Thus, where the name of the device has evolved (or changed) during the design process and its previous names appear in the supporting evidence document (for example), this historical information must be clearly specified in the introduction to the Technical Documentation. Omitting this information can potentially result in non-admissibility of the supporting evidence documents concerned (for example).
Similarly, the given names in the Technical Documentation for example to raw materials, packaging, must be consistent with those used in the supporting evidence document. Where this is not the case, the information must be explained in the Technical Documentation for the product reviewer to accept the document as effective.

c → Overall structure of the electronic file

i. Initial notification, renewal or modification file

In general terms, it is important to remember that the Technical Documentation must be structured in accordance with the framework of the GMED report and specific reports. These GMED documents differ depending on the type of medical devices (packaging, sterilizations, etc.).

In this context, it is recommended that the introduction to the Technical Documentation makes clear:

- **The purpose of submitting the file** and **the expected assessment scope** (especially if the Technical Documentation relates to several devices);
- An **explanatory letter specifying, for each of the points** (set out in Part 2 of this document), **the answers given and links to the corresponding Technical Documentation** or the rationale for the non-applicability or the non-impact of the modification on the specific point set out in the report.

ii. File for additional information request

On completion of the Technical Documentation assessment, there may be a number of outstanding questions that become the subject of Non-Conformity(ies) addressed to the manufacturer. In that case, the manufacturer is notified of the Non-Conformity(ies) and will receive a Non-Conformities form.
In return, the recommendation is to answer the Non-Conformity(ies) in the requested order. It will be expected to have for each Non-Conformity, the answers given and the links to the corresponding Technical Documentation (or the rationale for the non-applicability or the non-impact of the modification) specifically identified in a document. This document should be included in the introduction to the additional information file.

d → Technical Documentation content

The Technical Documentation includes **a table of contents covering every part of the Technical Documentation.** These tables of contents detail and specify (non-exhaustive):

- Document titles;
- Document number (reference, version and effective date) and the name and location of the electronic file;
- Chapter number;
- Annex number;
- The precise pagination;
- The table of contents for each annex.

3 The electronic file

a → Electronic files in PDF format

The electronic file includes multiple electronic documents. These documents must be **in Portable Document Format (PDF)**. It must be possible to conduct an active **(searchable)** word search within each PDF file for **the entirety of the documents**.

Filenames must provide a quick and clear link to the table of contents of the file. However, for technical reasons, **the path to the file is limited to 40 characters and must not contain any special characters** (@ _ - [], etc. are prohibited).

The recommendation is that the **diagrams, illustrations, and photos included in files remain in their original colors and resolutions**.

Lastly, the inclusion of several small files is preferable to a single file with a large number of pages.

b → Use of bookmarks and pagination

The PDF format offers **the use of bookmarks (within a tree structure)** as electronic dividers to structure the document and enable browsing on the basis of this organizational structure. It is essential to use these bookmarks for each document contained in the electronic file. Naturally, each bookmark should refer consistently to the electronic file table of contents (or the contents of the corresponding annex).
Lastly, the pagination, paragraph, chapter and document numbers will be shown on the header or footer of each page.

c → Submitting the electronic file to GMED

Each Technical Documentation must be sent in an electronic copy to GMED via a download platform of your choice.

The link to initiate the download is to be sent to the attention of your Project Manager and the Certification Assistant (Operations Coordinator at GMED NA).

For easy identification of the electronic file, please indicate the GMED project number in the subject of the email.

In case of difficulty to transmit your Technical Documentation in electronic format, please reach out your designated GMED contact.



II → INFORMATION TO BE PROVIDED IN THE CONTEXT OF ASSESSMENT

This list of information to be provided for the Technical Documentation assessment by GMED has been compiled in accordance with the requirements set out in Annex II to Regulation (EU) 2017/745.

As part of the assessment activities conducted by GMED, the Technical Documentation is subject to a preliminary review to verify that the Technical Documentation submitted is complete, and to identify, when necessary, any points where additional information is required before the assessment process begins.
The assessment is then planned based on the results of this review.

Note: Within the scope of products covered by Annex XVI "List of groups of products without an intended medical purpose referred to in Article 1(2)" to Regulation (EU) 2017/745, the term "medical device" shall be understood as the term "device".
Also, within the scope of products covered by Annex XVI, the term "consumer" is used. A "consumer" shall be understood as a natural person on whom a product without an intended medical purpose is intended to be used, as specified in Implementing Regulation (EU) 2022/2346.

Note: In the case where the Technical Documentation covers a device intended both for a medical and non-medical purpose, it is recommended to clearly identify and discriminate the elements concerning the medical and non-medical purpose, in particular in the following sections:

- Part I - General information:
  - Section 3: Trade name of the medical device

- Part II - Description and specification of the medical device, including variants and accessories:
  - Section 6: Class of the device and rationale
  - Section 8: Description of the claimed use

- Part III - Information to be provided by the manufacturer:
  - Section 12: Accompanying documents

- Part V - General safety and performance requirements:
  - Section 15: Requirements summary table in Annex I of Regulation (EU) 2017/745 (GSPR)

- Part VI - Benefit-risk analysis and risk management:
  - Section 16: Benefit-risk analysis referred to in Annex I of Regulation (EU) 2017/745
  - Section 17: Risk management file

- Part VII - Product verification and validation:
  - Section 18: Pre-clinical data: Demonstration that characteristics of the device conform with the standards, reference documents and/or applied state of the art in order to demonstrate the pre-clinical safety of the device and its compliance with the specifications, in particular:
    - Section 18.4: Mechanical, physical, chemical, electrical, etc. evaluation
    - Section 18.9: Usability engineering

- Part VIII - Post-market surveillance:
  - Section 19: Description of the provisions made for post-market surveillance of the device, including the post market clinical follow-up (PMCF) plan
  - Section 20: Post-market data

Note that the discrimination between the medical and non-medical purpose may also be necessary in other sections, depending on the characteristics of the device under assessment.

I – GENERAL INFORMATION	1	**LEGAL MANUFACTURER** Name, registered trade name or registered trademark and full address of the manufacturer as defined in the Regulation. **Note 1:** The legal structure responsible for placing the medical device on the market must be specifically and consistently stated throughout the Technical Documentation and the information provided by the manufacturer (instructions, user manual, labelling, brochure, catalog and similar publications, etc.).
	2	**AUTHORIZED REPRESENTATIVE IN THE EUROPEAN UNION (WHERE APPLICABLE)** Name or registered trade name and full address of the authorized representative in the European Union.
	3	**TRADE NAME OF THE MEDICAL DEVICE** 3.1. Identification of the device and its variants, and various configurations of the device, with their commercial references. 3.2. Identification of the Member States in which the product is intended to be sold or in which the product is already marketed.
	4	**EXISTING CERTIFICATION** 4.1. Identification of the EC quality system certificate covering the device (section 1 of Annex IX of Regulation (EU) 2017/745). 4.2. Identification of the certificate of conformity in section 2 of Annex IX of Regulation (EU) 2017/745, where applicable. 4.3. Has the device been the subject of a conformity assessment under another status or been registered outside the EU?

II INFORMATION TO BE PROVIDED IN THE CONTEXT OF ASSESSMENT (continued)

I – GENERAL INFORMATION

5

STATEMENTS

5.1. A statement indicating whether or not a product of animal origin is used in the device manufacturing.

See Technical memo: Material of Animal Origin - Information to be provided

Note 5.1.1: A derived product is a material or substance obtained from a tissue of animal origin by means of a manufacturing process (e.g., collagen, gelatin, enzymes, etc.).

Note 5.1.2: For example, the reagents of biological origin used in the medical device manufacturing process may be peptones of bovine origin, amino acids of porcine origin, murine monoclonal antibodies or fetal calf serum (used in cell cultures).

Note 5.1.3: Where the assessment relates to the modification of a medical device using a material of animal origin (or the introduction of a variant of an existing product or extension of the range), the absence of any impact on viral safety and reduction in Transmissible Spongiform Encephalopathy (TSE) risk (where applicable) as a result of the modification must be documented.

Note 5.1.4: Tallow derivatives are addressed in section 18.3 of this technical memo.

5.2. A statement indicating whether or not the device incorporates a substance likely to be considered as a medicinal product (with reference to Annex I section 12.1 of Regulation (EU) 2017/745).

See Technical memo: Medicinal substance with an ancillary action incorporated in a medical device - Information to be provided

5.3. A statement indicating whether or not the device incorporates a substance derived from human blood.

5.4. A statement indicating whether or not the device is composed of substances or combinations of substances intended to be introduced into the human body and absorbed by the human body or locally dispersed therein (with reference to Annex I, section 12.2 of Regulation (EU) 2017/745).

See Technical memo: Medical device composed of substances or combinations of substances intended to be introduced into the human body via a body orifice or applied to the skin and absorbed by or locally dispersed in the human body - Information to be provided

II INFORMATION TO BE PROVIDED IN THE CONTEXT OF ASSESSMENT (continued)

II - DESCRIPTION AND SPECIFICATION OF THE MEDICAL DEVICE, INCLUDING VARIANTS AND ACCESSORIES	6	**CLASS OF THE DEVICE AND RATIONALE** 6.1. Reasons why the product constitutes a medical device and/or reasons why the product constitutes a product without a medical purpose listed in Annex XVI of Regulation (EU) 2017/745. 6.2. Classification according to Annex VIII of Regulation (EU) 2017/745: rule applied and justification for the classification. **Note 6.2:** In the case of groups of certain active products without a medical purpose, refer to the reclassification rules of Implementing Regulation (EU) 2022/2347.
	7	**DESCRIPTION OF THE DEVICE** 7.1. Unique Device Identifier (UDI-DI). 7.2. EMDN (or GMDN) code. 7.3. Where applicable, a description of the accessories, other medical devices and products that are not medical devices which are intended to be used in combination with the device (including those **covered and not covered** by the assessment). 7.4. Where applicable, identification and description of the various configurations or variants of the device. 7.5. Where applicable, identification and description of the associated software program(s). 7.6. A description of the commercial presentation (the device in its packaging), including detailed photos.

II INFORMATION TO BE PROVIDED IN THE CONTEXT OF ASSESSMENT (continued)

II - DESCRIPTION AND SPECIFICATION OF THE MEDICAL DEVICE, INCLUDING VARIANTS AND ACCESSORIES

8

DESCRIPTION OF THE CLAIMED USE

8.1. Claimed medical purpose.

Note 8.1: In the case of products without a medical purpose listed in Annex XVI of Regulation (EU) 2017/745, the claimed purpose of the product is not medical.

8.2. Device's intended users and/or consumers*.

* See the associated definition in the note in the preamble of the part 2: Information to be provided in the context of assessment of this technical memo: term used within the scope of products covered by Annex XVI of Regulation (EU) 2017/745

8.3. Indications and intended conditions of use related to the disease, pathology and/or disability treated, the patients and/or users and/or consumers* target group and applicable patients and/or users and/or consumers* selection criteria.

Note 8.3: In the case of products without a medical purpose listed in Annex XVI of Regulation (EU) 2017/745, there are no indications.

8.4. Principle of operation of the device and its mode of action, including the use of accessories that are not assessed as part of this assessment, where applicable.

8.5. Clinical performances claimed for the intended use relative to the disease, pathology and/or disability treated.

Note 8.5.1: For joint replacements, please specify the functional performances claimed (e.g., minimum and maximum relative angular movements expected between the parts of the skeleton to which the joint replacement relates, assumed wear of the joint surfaces, etc.).

Note 8.5.2: In the case of products without a medical purpose listed in Annex XVI of Regulation (EU) 2017/745, clinical performances claims are limited to the ability of the product to achieve its intended purpose.

8.6. Claimed clinical benefits.

Note 8.6: In the case of products without a medical purpose listed in Annex XVI of Regulation (EU) 2017/745, the requirement to demonstrate a clinical benefit shall be understood as a requirement to demonstrate the performance of the product, as specified in section 9 of Article 61 of Regulation (EU) 2017/745.

8.7. Contra-indications, side-effects and warnings.

8.8. Lifetime of the device (the period over which it continues to deliver its clinical performances under normal conditions of use): depending on the medical device concerned, it is important to be precise, for example, by making a distinction between the period of time during which the device remains effective, and the length of time that the device remains in the human body but is no longer effective.

Note 8.8: In the case of products without a medical purpose listed in Annex XVI of Regulation (EU) 2017/745, clinical performances claims are limited to the ability of the product to achieve its intended purpose.

8.9. Expiry date.

II INFORMATION TO BE PROVIDED IN THE CONTEXT OF ASSESSMENT (continued)

II - DESCRIPTION AND SPECIFICATION OF THE MEDICAL DEVICE, INCLUDING VARIANTS AND ACCESSORIES

9

CHARACTERISTICS OF THE COMPONENTS, FINISHED PRODUCT (DEVICE) AND ACCESSORIES COVERED BY THE ASSESSMENT, WHERE APPLICABLE

9.1. General description of key functional elements, such as parts or components (formulation, composition and functionality).

Note 9.1: For medical devices containing particles smaller than 100 nm, the documentation provided by the manufacturer must provide characteristics such as: agglomeration/aggregation status, composition (e.g., chemical composition and structure), particle size and distribution, purity/impurity, shape, solubility (hydrophobicity, lipid solubility or water solubility), stability, surface area, surface chemistry, surface load, coating characteristics, etc.

9.2. Raw material specifications (see note 9.4.1): physical, chemical, biological, etc., together with their reference, revision number and revision date.

9.3. Specifications of additives, contaminants and residues from the manufacturing process (see note 9.4.1): physical, chemical, biological, etc. together with their reference, revision number and revision date.

9.4. Specifications of primary packaging (the packaging in direct contact with the medical device), together with its reference, revision number and revision date.

Note 9.4.1: Raw materials, additives, contaminants, residues from the manufacturing process and primary packaging items, where any change could potentially compromise compliance with the general safety and performance requirements.

Note 9.4.2: The list of specifications, required in the previous points, together with the name and address of the suppliers and distributors must be provided in the following format, as well as in a reproducible electronic format (Annex 1 of this technical memo).

Raw materials, additives, contaminants, residues from the manufacturing process and primary packaging items	Specifications (Physical, surface, chemical composition, size, etc. characteristics)	Supplier / Address	Distributor / Address

9.5. Identification of materials in contact with the patient (when implanted or not), and/or the user and/or the consumer*, regardless of whether contact with the human body is direct or indirect.

9.6. Specifications of the finished product: all technical specifications, such as device characteristics, dimensions and all variants/configurations.

9.7. Where applicable, all technical specifications of accessories, including accessory characteristics, dimensions and all variants/configurations.

9.8. Where applicable, plans and diagrams together with their references, revision number and revision dates and a photograph.

II INFORMATION TO BE PROVIDED IN THE CONTEXT OF ASSESSMENT (continued)

II - DESCRIPTION AND SPECIFICATION OF THE MEDICAL DEVICE, INCLUDING VARIANTS AND ACCESSORIES

10	**REFERENCE TO PREVIOUS AND SIMILAR GENERATIONS OF THE DEVICE**
	10.1. Changes in a medical device already CE marked by the manufacturer (provide a general overview of the previous generation(s) of the device produced by the manufacturer, if any).
	10.2. Development basis of similar medical device from a competitor (provide a general overview of similar devices identified available on the EU or international markets, if any).
	10.3. Totally new medical device.
	10.4. Explanation of any new features.
	Whether the device is innovative or not must be specified and further explained if applicable. To do so, the degree to which the device is innovative must be defined on the basis of the criteria given in the Single Information Form, FI-INF tab.
11	**HISTORY OF CHANGES (NOT APPLICABLE IN CASE OF INITIAL ASSESSMENT)**
	11.1. Changes made to the device since its initial market introduction (specify the date of market introduction).
	11.2. Identification of modifications not yet notified.
	11.3. Regulatory, standards or state of the art evolution.

II INFORMATION TO BE PROVIDED IN THE CONTEXT OF ASSESSMENT (continued)

III – INFORMATION TO BE PROVIDED BY THE MANUFACTURER

12

ACCOMPANYING DOCUMENTS

Note 12: In the case of products without a medical purpose listed in Annex XVI of Regulation (EU) 2017/745, information supplied with the product shall not bear any clinical benefit claim or statement. Refer to the applicable annexes of Implementing Regulation (EU) 2022/2346 which list the content of the specific instructions for use and label for these products.

12.1. Instructions for use:

- Content of the instructions for use or user manual (or draft manual) specific to the device in the languages accepted in the Member States where the device is intended to be sold;
- Where applicable, another (other) document(s) for patients or users or consumers*, where this (or these) is (or are) referred to in the instructions for use (e.g., surgical technique, implant card, brochure, etc.)

Note 12.1.1: When instructions for use in electronic form is implemented, conformity with Implementing Regulation (EU) 2021/2226 will be assessed. In this context, the Technical Documentation shall include the impact analysis of the instructions for use in electronic form, especially in relation to the following information:
- Characteristics of the components, finished product (device) and accessories covered by the assessment, where applicable (see section 9);
- Description of the accompanying documents;
- Demonstration of conformity with applicable requirements of the Implementing Regulation (EU) 2021/2226 shall be documented. It is recommended to use a summary table format;
- Risk management file (see section 17);
- Usability engineering (see section 18.9);
- Impact analysis on the special processes validation (e.g., packaging, sterilization processes, etc.);
- Requirements summary table in Annex I of Regulation (EU) 2017/745 (GSPR) (see section 15).

Note 12.1.2: Please note that the Implementing Regulation (EU) 2021/2226 also establishes certain requirements concerning contents and web presentation for instructions for use that are provided in electronic form in addition to instructions for use in paper form.

12.2. Labelling:

- Label(s) on the device and on its packaging, such as single unit packaging, sales packaging and/or transportation packaging in case of specific management conditions, in the languages accepted in the Member States where the device is intended to be sold;
- The text and graphic elements of the commercial packaging (mockup), in the languages accepted in the Member States where the device is intended to be sold.

Note 12.2: Refer to the requirements of Annex I section 23 of Regulation (EU) 2017/745, product standards, EN ISO 15223-1, etc.

12.3. Other accompanying documents (e.g., surgical technique, user manual, maintenance manual, calibration manual, brochure, catalog and similar publications, etc.).

II INFORMATION TO BE PROVIDED IN THE CONTEXT OF ASSESSMENT (continued)

IV – DESIGN AND MANUFACTURING INFORMATION

13 DESIGN INFORMATION (IF APPLICABLE)

13.1. Planning the design project (final version) making clear the stages of the device design.

13.2. Identification of design verification outputs.

13.3. Records of the results and conclusions of the design and development transfer.

13.4. Reports of all design reviews conducted during the project implementation.

14 MANUFACTURING INFORMATION

14.1. Name and address of suppliers/distributors of raw materials, additives, contaminants, residues from the manufacturing process and primary packaging items.

Note 14.1.1: These are the main components, raw materials, additives, contaminants, residues from the manufacturing process, and primary packaging items, where any change could potentially compromise compliance with the general safety and performance requirements.

Note 14.1.2: The identification of suppliers and distributors is to be transmitted (Annex 1 of this technical memo).

14.2. Manufacturing flowchart (identifying the processes implemented and specifying whether they are validated or verified, together with the in-process and final controls performed), including where these stages are subcontracted.

Note 14.2.1: This document must also be provided in a reproducible electronic format (Annex 2 of this technical memo).

Note 14.2.2: Any intermediate cleaning stage(s) must also be specified, if applicable.

14.3. Description of the manufacturing methods and controls performed, including where these stages are subcontracted.

Note 14.3: Description must be provided of the in-process controls and controls applied to the final product. The control criteria on the critical characteristics of the device, including where these are subcontracted, must also be provided.

14.4. A description of the validated manufacturing process(es) and validation report(s) (OQ and PQ), including where these processes are subcontracted.

Note 14.4.1: This must at least identify the following information:
- Description of the validated process with the precise identification of the equipment concerned;
- Identification of associated validation reports (OQ/PQ) with their reference, revision number and revision date;
- Identification of critical process parameters as well as validated tolerance intervals (mini / maxi).

Note 14.4.2: The list of processes must also be transmitted in a reproducible electronic format (Annex 3 of this technical memo).

14.5. Description of the manufacturing environment(s), together with its (or their) corresponding classification.

14.6. Identification of the subcontractors' manufacturing sites, and QMS certifications (if any), certification bodies and validity of the current certificates, if any.

Where a process has been the subject of a previous GMED assessment in the context of Regulation (EU) 2017/745, in a Masterfile format (validation of a process covering several devices covered by different Technical Documentations and/or dependent on different categories and/or generic groups):

14.6.1. Identification of the process(es) concerned.

14.6.2. Identification of the number and date of GMED assessment report, ruling favorably.

14.6.3. Rationale relating to the inclusion of the device, subject of the assessment, in the validation of the process already assessed (inclusion of the product in a defined family without challenging the worst-case).

II INFORMATION TO BE PROVIDED IN THE CONTEXT OF ASSESSMENT (continued)

V – GENERAL SAFETY AND PERFORMANCE REQUIREMENTS	15	**REQUIREMENTS SUMMARY TABLE IN ANNEX I OF REGULATION (EU) 2017/745 (GSPR)** 15.1. General safety and performance requirements that apply to the device and an explanation as to why others do not apply. 15.2. Method(s) used to demonstrate conformity with each applicable general safety and performance requirements. 15.3. Harmonized standards, common specifications or other solutions applied. **Note 15.3:** The list of standards and documents applied must also be provided in a reproducible electronic format (Annex 4 of this technical memo). 15.4. Precise identification of the controlled documents providing evidence of conformity to each harmonized standard, common specification or other method applied to demonstrate conformity with the general safety and performance requirements. The location of controlled documents within the Technical Documentation and, where applicable, the Technical Documentation summary is to be provided.
VI – BENEFIT-RISK ANALYSIS AND RISK MANAGEMENT	16	**BENEFIT-RISK ANALYSIS REFERRED TO IN ANNEX I OF REGULATION (EU) 2017/745** The manufacturer's benefit-risk analysis, referred to in sections 1 and 8 of Annex I of Regulation (EU) 2017/745, is conducted using data from the clinical data evaluation, PSUR and PMS conclusions and risk management outcomes. In the case of products without a medical purpose listed in Annex XVI of Regulation (EU) 2017/745, the general safety requirements set out in sections 1 and 8 of Annex I of Regulation (EU) 2017/745 shall be understood to mean that the product, when used under the conditions and for the purposes intended, does not present a risk at all or presents a risk that is no more than the maximum acceptable risk related to the product's use which is consistent with a high level of protection for the safety and health of persons.
	17	**RISK MANAGEMENT FILE** 17.1. Description of the methods used for risk analysis, risk control and identification of the relevant standard (e.g., NF EN ISO 14971, current version). 17.2. Definition of the lifecycle of the device: design, ..., aging, disposal. 17.3. Complete risk management file including (in accordance with EN ISO 14971, current version): 17.3.1 Risk management plan. 17.3.2 Summary of the steps in risk management: analysis, risk assessment, implementation and verification of risk control measures, assessment of the acceptability of any residual risk including overall. 17.3.3 Report (review report of the risk management process). **Note 17.1:** ∘ The risks associated with all the phases of the medical device lifecycle (design, manufacture, utilization, aging, disposal) are to be evaluated and controlled; ∘ It is important to clearly identify, document and justify the decision-making criteria whose application has resulted in demonstrating that the risks identified are acceptable in light of the anticipated benefits to patients; ∘ All the known and foreseeable hazards and hazardous situations (combinations or sequences of events) must be identified; ∘ The level of acceptability of the risk under specific conditions must be considered with reference to the "generally recognized state of the art"; ∘ Residual risks and their processing operations must be identified. A decision must be made regarding the acceptability of overall residual risk. Mention should be made where the medical benefits outweigh the residual risk. **Note 17.2:** In the case of products without a medical purpose listed in Annex XVI of Regulation (EU) 2017/745, refer to the risk management requirements defined in Implementing Regulation (EU) 2022/2346.

II INFORMATION TO BE PROVIDED IN THE CONTEXT OF ASSESSMENT (continued)

VII – PRODUCT VERIFICATION AND VALIDATION

18	**PRE-CLINICAL DATA: DEMONSTRATION THAT CHARACTERISTICS OF THE MEDICAL DEVICE CONFORM WITH THE STANDARDS, REFERENCE DOCUMENTS AND/OR APPLIED STATE OF THE ART IN ORDER TO DEMONSTRATE THE PRE-CLINICAL SAFETY OF THE DEVICE AND ITS COMPLIANCE WITH THE SPECIFICATIONS**
18.1	**CONSIDERATIONS FOR THE USE OF ALTERNATIVE MATERIALS OR SUBSTANCES** • Considerations for the use of alternative materials or substances; • Rationale behind the choice of materials or substances relative to the available alternatives.
18.2	**PRESENCE WITHIN THE DEVICE OF SUBSTANCES IN A CONCENTRATION GREATER THAN 0,1% WEIGHT BY WEIGHT (W/W) (WHERE APPLICABLE)** Information regarding whether the device(s), subject of this assessment, or parts of devices or materials used are: • Invasive and come into direct contact with the human body, or; • Intended to (re)administer medicines, body fluids or other substances, including gases, to/from the body, or; • Intended to transport or store medicines, body fluids or substances, including gases, to be (re)administered to the body. If yes to at least one of the previous questions: • Indicate whether substances which are carcinogenic, mutagenic or toxic to reproduction of category 1A and 1B or substances having endocrine disrupting properties (as referred to in section 10.4.1 of Annex I of Regulation (EU) 2017/745) are present in a concentration greater than 0,1% weight by weight/Identification of the substance(s) concerned and quantity present; • Justification relating to the presence of carcinogenic, mutagenic or toxic to reproduction and/or endocrine disrupting substances. This justification is based on the following elements: - An analysis and estimation of patient or user or consumer* potential exposure to the substance, accompanied by the identification of the weight by weight concentration; - An analysis of possible alternative substances, materials or designs, including information about independent research, peer-reviewed studies, and scientific opinions from relevant scientific committees if available and an analysis of the availability of such alternatives; - Argumentation as to why possible alternative substances and/or materials, if available, or design changes, if feasible, are inappropriate in relation to maintaining the functionality, performance and the benefit-risk ratio of the product; including taking into account if the intended use of the devices includes treatment of children or treatment of pregnant or breastfeeding women or treatment of other patient groups considered particularly vulnerable to such substances and/or materials; - Where applicable and available, the latest relevant scientific committee guidelines in accordance with sections 10.4.3. and 10.4.4. of Regulation (EU) 2017/745. **Note 18.2:** In the case of products without a medical purpose listed in Annex XVI of Regulation (EU) 2017/745, the term "benefit-risk ratio" is to be replaced by "risk".

II INFORMATION TO BE PROVIDED IN THE CONTEXT OF ASSESSMENT (continued)

VII – PRODUCT VERIFICATION AND VALIDATION

18.3

PRESENCE OF ANIMAL TALLOW DERIVATIVES

Information regarding whether the device(s), subject of this assessment, or parts of devices or materials used are intended to come into contact with the human body or non-intact skin.

If yes:
- Indicate the presence of animal tallow derivatives that may have been used particularly in the processing of plastics and/or packaging materials for these medical devices/Identification of the substance(s) concerned.

Note 18.3.1: Tallow is fat from animal tissue, including subcutaneous, abdominal, and intermuscular areas and bones.

- List of components and raw materials whose manufacturing processes involve the use of materials of animal origin, together with the identification of their suppliers (specific information regarding contact or not with the patient and/or user and/or consumer* is also required);
- Copies of supplier certificates, including information relating to the parameters applied to the manufacturing of tallow derivatives, to enable assessment of whether or not Regulation (EU) No 722/2012 applies (i.e., identification of the plastics and/or packaging materials concerned, and the conditions under which tallow derivatives have been manufactured);
- Documented rationale regarding the applicability or not of Regulation (EU) No 722/2012 on the basis of certificates provided by suppliers.

Note 18.3.2: Additional regulatory proceedings may need to be carried out in the event that Regulation (EU) No 722/2012 becomes applicable.

II INFORMATION TO BE PROVIDED IN THE CONTEXT OF ASSESSMENT (continued)

VII – PRODUCT VERIFICATION AND VALIDATION

18.4

MECHANICAL, PHYSICAL, CHEMICAL, ELECTRICAL, ETC. EVALUATION

Reports relating to the tests carried out, accompanied by the certificate of the laboratory that carried out the tests (e.g., 17025 accreditation) and justification in the absence of pre-clinical testing.

Note 18.4.1: When the device is designed to be used sterile, the testing should be performed on the sterile device.

Furthermore, when an assessment refers to an assessment report or company document that is more than 5 years old, the corresponding data must be transmitted and be the subject of a justification of its applicability.

Note 18.4.2: The results should preferably be presented in the form of a table. Example below:

Standard / Document / State of the art	Characteristic / Specification	Requirements / Acceptance criteria	Method used to demonstrate compliance with the requirement (testing and rationale)	Identification of the laboratory (name and address) Identification of the accreditation and expiry date	Report reference	Worst case justification	Compliance outcome

Definitions:
- Characteristic: intrinsic property of the medical device
- Specification: document stating the requirements applying to a characteristic
- Requirement: expressed expectation, whether implied or mandatory
- Compliance: fulfillment of a requirement

Note 18.4.3: Where tests have not been conducted in accordance with the provisions set out in the applicable standards or reference documents, the manufacturer must document the demonstration of equivalence of the solutions adopted.

Note 18.4.4: Where calculations (including finite element analysis) and simulations fail to demonstrate achievement of the performances claimed by the manufacturer, and, where applicable, the compliance of the device with the applicable standards or reference documents, the manufacturer must consider the necessity to confirm these simulations and calculations by means of testing.

Note 18.4.5: Justification must be provided where no pre-clinical data is available. Furthermore, when an assessment refers to an assessment report or company document that is more than 5 years old, the corresponding data must be transmitted and be the subject of a justification of its applicability.

Note 18.4.6: The safety of devices emitting radiation and electrical devices in relation to these characteristics must be taken into account.

Note 18.4.7: Where studies have been conducted using animal models, detailed information must be provided, specifically regarding the purpose of research, the methodology, results, analyses, and conclusions, accompanied by the rationale and limitations of the selection of the model(s).

Note 18.4.8: Where simulated use testing has been conducted, the manufacturer must provide detailed information.

Note 18.4.9: Where the manufacturer conducts pre-clinical tests in-house or in a laboratory without 17025 accreditation, GMED reserves the right to request the corresponding validations of the implemented testing methods.

Note 18.4.10: Where an EU type-examination, this section should be completed jointly with the EU type-examination report.

Note 18.4.11: In the case of products without a medical purpose listed in Annex XVI of Regulation (EU) 2017/745, the verification of specific risk control measures listed in the applicable annexes of Implementing Regulation (EU) 2022/2346 and relating to the requirements concerning the product design are to be considered.

II INFORMATION TO BE PROVIDED IN THE CONTEXT OF ASSESSMENT (continued)

VII – PRODUCT VERIFICATION AND VALIDATION	18.5	**WHERE DEVICES WITH A MEASURING FUNCTION ARE PLACED ON THE MARKET** Description of the methods used to ensure the accuracy indicated in the specifications.
	18.6	**IF THE DEVICE IS TO BE CONNECTED TO ANOTHER (OTHER) DEVICE(S) IN ORDER TO OPERATE AS INTENDED** 18.6.1. A description of this connection/configuration must be provided. 18.6.2. This description must include evidence that it meets the general safey and performance requirements for all affected devices when connected, with regards to the specifications indicated by the manufacturer.
	18.7	**DEMONSTRATION THAT CHARACTERISTICS ARE MAINTAINED** 18.7.1. As part of the stability study to justify the shelf life of the device (taking into account the device lifecycles). 18.7.2. During the expected device lifetime. 18.7.3. Validation of the device reprocessing, where applicable.
	18.8	**CONCLUSION OF THE MANUFACTURER** Conclusion of the manufacturer taking into account considerations regarding the use of other materials or substances.
	18.9	**USABILITY ENGINEERING** Report on usability, including the identification of the reference standard (e.g., IEC 62366-1, current version). **Note 18.9:** In the case of products intended both for a medical and non-medical purpose, a usability engineering for the medical and non-medical purpose is to be issued for each purpose.

II INFORMATION TO BE PROVIDED IN THE CONTEXT OF ASSESSMENT (continued)

VIII – POST-MARKET SURVEILLANCE

19 DESCRIPTION OF THE PROVISIONS MADE FOR POST-MARKET SURVEILLANCE OF THE DEVICE, INCLUDING THE POST-MARKET CLINICAL FOLLOW-UP (PMCF) PLAN

Note 19: In the case of products intended both for a medical and non-medical purpose, the description of the provisions made for post-market surveillance of the device, including the post-market clinical follow-up (PMCF) plan, for the medical and non-medical purpose is to be issued for each purpose.

19.1. Post-market surveillance plan in accordance with Annex III section 1.1 of Regulation (EU) 2017/745.

19.2. Post- Market Clinical Follow-up (PMCF) plan.

19.3. Periodic Safety Update Report (PSUR) in accordance with Article 86 of Regulation (EU) 2017/745.

19.4. Post-market surveillance report in accordance with Article 85 of Regulation (EU) 2017/745.

19.5. PMCF evaluation report.

19.6. Changes in provisions and reasons for changing the provisions.

20 POST-MARKET DATA

Note 20: In the case of products intended both for a medical and non-medical purpose, the post-market data for the medical and non-medical purpose is to be issued for each purpose.

20.1. Date of placing on the market.

20.2. Number of products sold inside and outside Europe.

20.3. Estimated population using the device (population size and characteristics).

20.4. Number and type of complaints about technical characteristics, clinical performance and usability of the device received since the initial market placement or CE marking.

Note 20.4: In the case of products without a medical purpose listed in Annex XVI of Regulation (EU) 2017/745, clinical performances claims are limited to the ability of the product to achieve its intended purpose.

20.5. Number and type of medical device vigilance reports received since the initial market placement or CE marking.

20.6. Data from clinical studies, including post market clinical follow-up and registries.

20.7. Modifications made and/or corrective actions taken following incidents reported and potential revision of the risk management file.

II INFORMATION TO BE PROVIDED IN THE CONTEXT OF ASSESSMENT (continued)

MANDATORY ANNEXES

1 IDENTIFICATION OF RAW MATERIALS, ADDITIVES, CONTAMINANTS, RESIDUES FROM THE MANUFACTURING PROCESS, AND PRIMARY PACKAGING ITEMS

1. Raw material specifications (see note): physical, chemical, biological, etc., together with their reference, revision number and revision date.
2. Specifications of additives, contaminants, and residues from the manufacturing process (see note): physical, chemical, biological, etc., together with their reference, revision number and revision date.
3. Specifications of primary packaging (the packaging in direct contact with the medical device) (see note), together with its reference, revision number and revision date.

Note: Raw materials, additives, contaminants, residues from the manufacturing process, and primary packaging items, where any changes could potentially compromise compliance with the general safety and performance requirements.

Note: The list of specifications required in the previous points together with the name and address of the suppliers and distributors must be provided in the following format, in a reproducible electronic format:

Raw materials, additives, contaminants, residues from the manufacturing process, and primary packaging items	Specifications (Physical, surface, chemical composition, size, etc. characteristics)	Supplier / Address	Distributor / Address

2 MANUFACTURING PROCESSES AND CONTROLS FLOWCHART

Manufacturing flowchart identifying the processes implemented and specifying whether they are validated or verified, together with the in-process and final controls performed, including where these stages are subcontracted.

3 IDENTIFICATION OF VALIDATED PROCESSES

The list of validated processes must be transmitted. This must at least identify the following information:

1. Description of the validated process with the precise identification of the equipment concerned;
2. Identification of associated validation reports (OQ/PQ) with their reference, revision number, and revision date;
3. Identification of critical process parameters as well as validated tolerance intervals (mini/maxi).

Note: The list of validated processes is to be transmitted in the following format, also in a reproducible electronic format:

Description of the validated process	Equipment identification	Validation reports (OQ/PQ) (ref. / rev. / date)	Critical parameters and validated tolerance intervals (mini/maxi)
Process	*Equipment*	*Report ref. rev. date*	*Critical parameter 1 ± X*
			Critical parameter 2 ± X
			Critical parameter 3 ± X

LIST OF STANDARDS AND DOCUMENTS APPLIED

List of common specifications (e.g., Implementing Regulation (EU) 2022/2346 for products without a medical purpose listed in Annex XVI of Regulation (EU) 2017/745), standards, whether harmonized or non-harmonized, and other documents (e.g., European Pharmacopoeia monographs) applied.

Where harmonized standards are not used, the solutions implemented must be equivalent and the equivalence documented.

Note: This document is also to be provided in a reproducible electronic format.

II INFORMATION TO BE PROVIDED IN THE CONTEXT OF ASSESSMENT (continued)

ADDITIONAL ANNEXES

1	CLINICAL DATA EVALUATION
	See Guide: Clinical Evaluation - Summary of safety and clinical performance
2	CLINICAL DATA EVALUATION FOR PRODUCTS WITHOUT A MEDICAL PURPOSE (ANNEX XVI OF REGULATION (EU) 2017/745)
	See Guide: Clinical Evaluation - Summary of safety and clinical performance
3	BIOLOGICAL EVALUATION
	See Guide: Biological assessment of medical devices according to the ISO 10993-1 standard *See Technical memo: Biological evaluation report – Information to be provided*
4	PROCESS VALIDATION: CLEANING AND DISINFECTION
	See Technical memo: Cleaning process - Information to be provided
5	PROCESS VALIDATION: PACKAGING
	See Technical memo: Packaging process for terminally sterilized medical devices – Information to be provided
6	PROCESS VALIDATION: STERILIZATION
	See Technical memo: Irradiation sterilization process – Information to be provided *See Technical memo: Ethylene oxide sterilization process – Information to be provided* *See Technical memo: Moist heat sterilization process – Information to be provided*
7	USE OF MATERIALS OF ANIMAL ORIGIN
	See Technical memo: Material of Animal Origin - Information to be provided
8	DEVICE INCORPORATING A MEDICINAL PRODUCT
	See Technical memo: Medicinal substance with an ancillary action incorporated in a medical device - Information to be provided
9	SUBSTANCES OR COMBINATIONS OF SUBSTANCES ABSORBED BY, OR LOCALLY DISPERSED IN, THE HUMAN BODY
	See Technical memo: Medical device composed of substances or combinations of substances intended to be introduced into the human body via a body orifice or applied to the skin and absorbed by or locally dispersed in the human body - Information to be provided
10	SOFTWARE VALIDATION(S)
	See Technical memo: Software development and validation - Information to be provided
11	TYPE-EXAMINATION ASSESSMENT REPORT AND ASSOCIATED TEST PROGRAM
12	EXPERT REPORT(S)
13	NOTIFIED BODY ASSESSMENT REPORT PERIODIC SAFETY UPDATE REPORT (PSUR)

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